

Global Energy Storage

Global Energy Storage: Data center battery intensity rises as sentiment cools



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Data centers are becoming more battery intense with average duration increasing to c. 5hrs as more are built behind the meter. Batteries are used in data centers for centralized power (UPS), distributed power (BBU) and energy storage (ESS). While UPS and BBU demand is relatively short duration (<0.25hrs), ESS demand can be longer. With 24% of data centers under construction off grid and 39% of the pipeline off grid, battery energy storage is rising in duration with some hyperscalers having up to 8hours storage.

Assuming 103GW of data center capacity by 2030, cumulative battery capacity will rise to 277GWh. We assume US data center capacity will increase from 41GW to 103GW by 2030. We expect annual data center battery demand to increase 10x from 5GWh to 66GWh. Cumulative ESS demand for data centers reached 11GWh in 2025 and will likely increase to 277GWh by 2030. This implies the battery /data center capacity GWh/GW multiplier will rise from 0.3 to 2.7x. Newer projects have a multiplier of 4-8x, implying further scope for battery demand to increase.

Increased data center demand raises our total battery demand for the US to 486GWh by 2030 which represents 22% CAGR and a 15% increase on previous estimates. 70% of incremental demand is forecast to be from ESS which will likely grow from 49GWh (2025) to 266GWh by 2030. ESS capacity in the US is currently limited at 36GWh but set to grow to 102GWh this year as more EV plants are converted to ESS.

But while our base case has increased, it's also fair to say that so too has uncertainty in long term growth. Increased data center capex and the success of lower cost Chinese models is creating near term uncertainty in the pace of AI data center build out. While the longer term outlook looks positive, there is a risk that capex could slow if hyperscalers become less confident in returns from the AI business model. Upcoming mid-year results from the hyperscalers may provide further guidance.

Profitability remains another key question, with margins (ex-subsidy) close to breakeven. While demand looks set to grow by 20-25% CAGR though to 2030, margins excluding tax benefits remain negative or close to zero. While guidance is for Korean battery maker margins to turn positive in 2H26, this remains still the subject of debate. While we think that higher ESS utilization will raise margins, continued de-rating of EV demand in the US YTD (-25%) may continue to put pressure on margins in the near term.

Korean battery makers valuations have a better risk-reward (12x EV/EBITDA 2027 and 5x for 2030) but we would wait for a better entry point. LGES and SDI are pricing in 22-26% revenue CAGR to end of decade (in line with industry growth) and a return to high single digit margins. Longer term we assume mid to high single digit growth and margins, which give c. 10% & 30% upside from current levels for LGES and SDI respectively. While these stocks look more attractive, we would wait for a better entry point.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	16 Jul 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
373220.KS (LGES)	M	KRW	332,000	351,000	(29.1)%	KRW (4,580.94)	(2,364.33)	10,225		(72.5)	(140.4)	32.5
OLD				347,000				1,811.04	9,452.97			
051910.KS (LG Chem)	M	KRW	260,000	312,000	(34.9)%	KRW (13,258.70)	2,693.22	28,743		(19.6)	96.5	9.0
OLD				298,000				3,030.56	27,968			
006400.KS (SDI)	M	KRW	430,000	520,000	92.9%	KRW (9,933.80)	1,751.66	18,577		(43.3)	245.5	23.1
OLD								2,099.51	18,376			
ASIA			1,865.18									

PRICE TARGET CHANGE / ESTIMATE CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Korean battery makers look well placed to benefit from the growth in ESS demand in the US and the tariff wall & subsidies which have been put in place to protect them from Chinese competition. Data center demand in particular could be a material driver of growth with a 10x increase in cumulative demand through to the end of the decade. Even by 2030, the storage/capacity multiplier of 2.7x could well be below the 4-8x multiplier for new data centers which are increasingly being built off grid. With LFP capacity set to expand significantly this year, Korean battery makers also look better placed to service this demand. SDI is more exposed to the ESS trend than LGES, with almost half of battery sales set to come from ESS and relatively lower EV exposure. Valuations more attractive given the sell-off driving forward EV/EBITDA multiples to 12x and long term (2030) multiples to 5x. But with questions surrounding the sustainability of AI capex and continued uncertainty on 2H26 margins we still believe there are better entry points ahead. We remain Market-Perform on LGES, LG Chem and SDI.

Model updates: we make a series of mostly minor updates based on our recent demand update. For the details of the model updates, please see page 14 onwards.

VALUATION COMPS TABLE

Company	Price 20-Jul	Curcy	EV USD mn	Mkt cap USD mn	24A	25A	26E	27E	24A	25A	26E	27E	Sales Growth 24-25
LGES	319,500.0	KRW	69,432	50,510	-109.9	-101.9	-2,094.1	45.3	3.9	4.3	3.5	2.8	-8%
CATL(H)	616.0	HKD	350,197	368,912	0.0	37.4	26.0	21.3	0.0	5.7	3.9	3.2	19%
CATL(A)	372.5	CNY	240,705	259,421	36.5	26.0	18.3	14.8	4.7	3.9	2.7	2.2	19%
Samsung SDI	403,500.0	KRW	30,136	22,002	31.9	-60.9	80.5	23.7	2.5	3.3	2.8	2.3	-26%
Panasonic	1,607	JPY	23,722	60,363	20.9	27.1	38.3	21.5	0.4	0.4	0.5	0.5	7%
BYD	89.9	HKD	130,640	117,066	21.4	22.7	20.0	15.7	1.2	1.1	1.0	0.9	15%
SK Innovation	117,600	KRW	40,697	13,452	-14.2	-15.4	12.7	19.6	0.8	0.7	0.7	0.7	8%
Gotion High Tech	26.0	CNY	15,265	6,968	59.4	15.9	25.4	17.3	3.0	2.3	1.7	1.3	29%
EVE	50.6	CNY	20,318	16,222	27.5	24.7	15.8	11.9	3.0	2.1	1.4	1.1	41%
Sunwoda	16.2	CNY	8,748	4,418	19.2	15.0	12.6	8.7	1.1	0.9	0.7	0.6	25%
CALB	17.2	HKD	15,446	3,897	46.2	21.0	10.6	7.0	3.5	2.6	1.5	1.2	38%

Company	LTM %	Rel LTM %	LTM High	LTM Low	GPM (%) 25A	OPM (%) 25A	ND/E 24A	25A	24A	25A	26E	27E	25A EV/Capacity (USD M/GWh)
LGES	-1%	-55%	437,500	247,000	18%	6%	40%	48%	28.9	20.5	18.4	12.3	196
CATL(H)	64%	56%	795	392	25%	19%	-34%	-46%	0.0	24.1	17.1	14.0	467
CATL(A)	38%	-7%	469	255	25%	19%	-34%	-46%	20.4	16.6	11.7	9.6	321
Samsung SDI	144%	8%	820,000	180,100	8%	-13%	42%	34%	16.3	89.4	18.9	11.7	479
Panasonic	182%	102%	4,770	1,411	30%	5%	11%	15%	4.5	4.1	5.3	3.7	395
BYD	-28%	-29%	136	71	18%	4%	-51%	5%	9.3	9.0	7.1	6.0	278
SK Innovation	7%	-49%	166,000	87,700	6%	1%	78%	68%	20.3	16.2	8.5	9.1	369
Gotion High Tech	-10%	-25%	49.8	24.3	18%	7%	125%	122%	26.3	16.9	13.8	11.4	70
EVE	16%	-26%	94.4	43.4	17%	8%	37%	49%	22.0	17.0	12.2	9.7	157
Sunwoda	-21%	-49%	37.9	16.0	16%	3%	61%	99%	16.0	14.7	9.0	6.9	107
CALB	1%	-9%	39.3	16.3	17%	6%	96%	106%	26.7	19.6	10.6	8.0	82

CATL, LG Energy Solution, and Samsung SDI are covered by Neil Beveridge. BYD is covered by Eunice Lee. All other stocks are not covered by Bernstein.

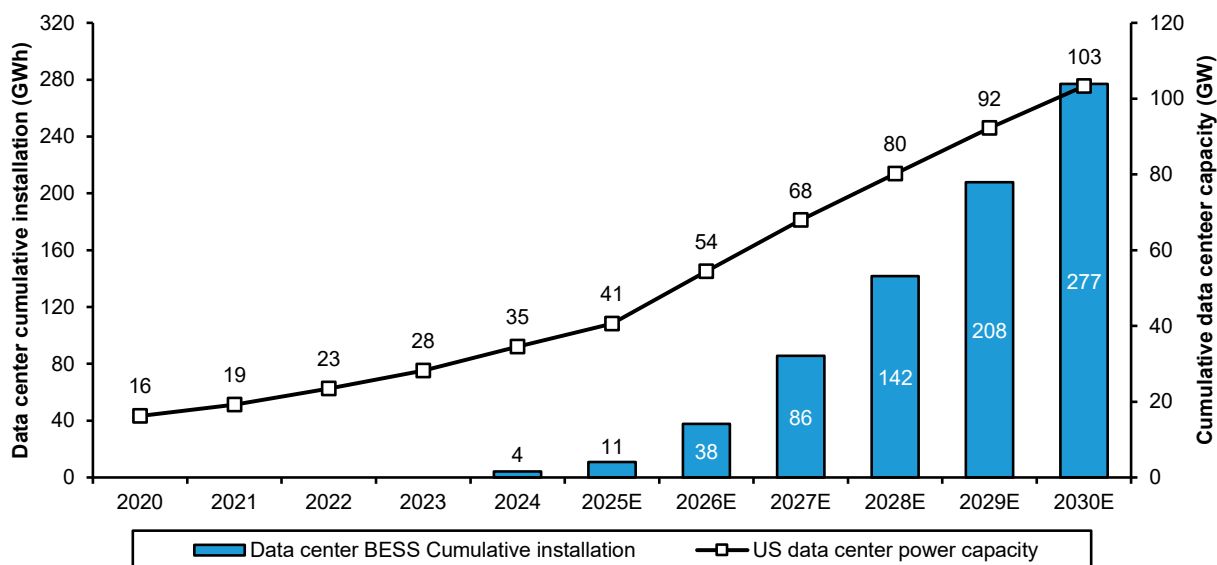
Source: Bloomberg (consensus), Bernstein analysis

DETAILS

DATA CENTER DEMAND IN US

We expect cumulative U.S. data center power capacity to reach 103 GW by 2030, driving incremental battery demand to 69 GWh in 2030 and cumulative battery installations to 277 GWh.

EXHIBIT 2: **Total capacity is expected to reach 103 GW in 2030, with cumulative battery capacity reaching 277 GWh.**



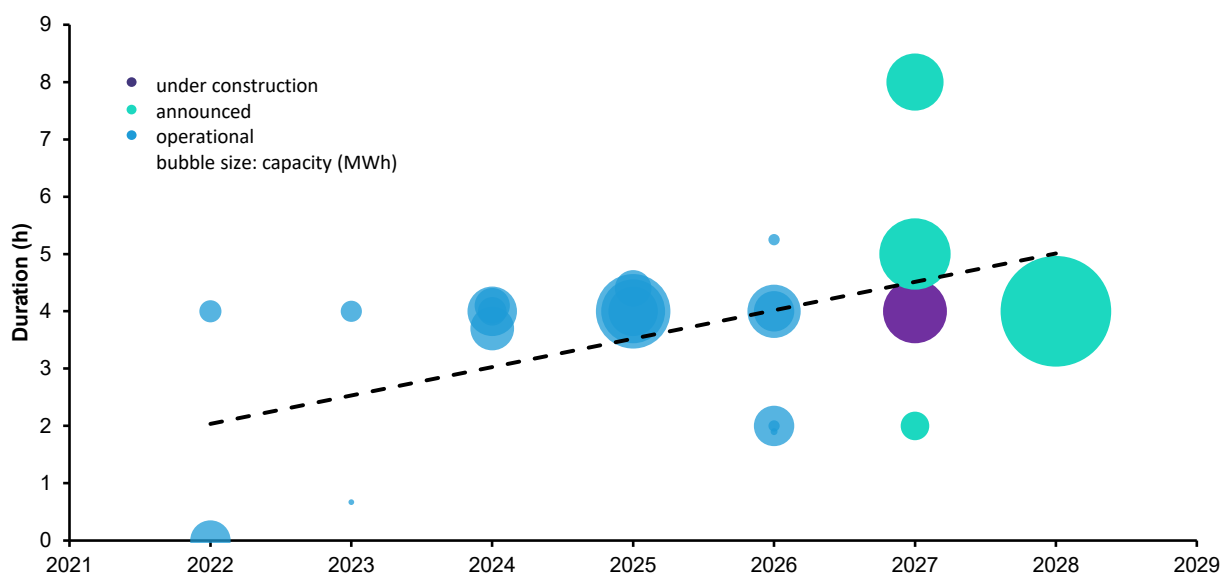
Source: BNEF, Bernstein analysis and estimates

Based on more than 20 announced U.S. data center projects, we have compiled a project-level summary table. The table shows that most projects are structured as utility-scale DC-PPAs or on-site behind-the-meter installations, with both power capacity and battery demand increasing clearly toward the end of the decade. For projects announced after 2024, the implied battery duration is generally above four hours, typically in the 4–8 hour range. This suggests that new data centers are increasingly adopting longer-duration battery solutions.

EXHIBIT 3: **Project-Level Benchmark for US Data Center BESS**

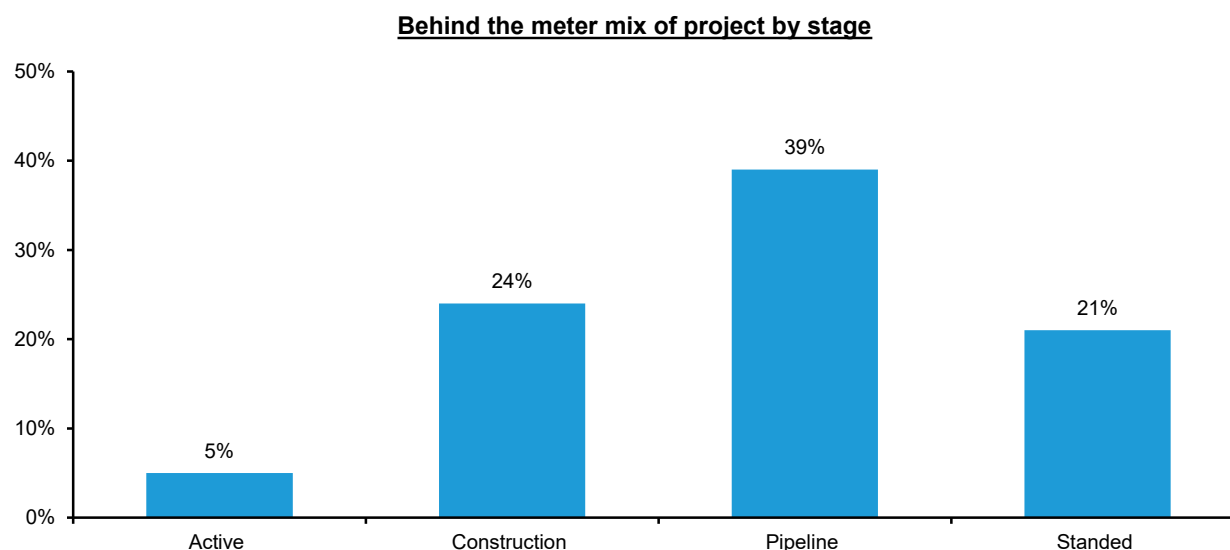
Project Name	Company / Operator	Year (Op./Ann.)	Power (MW)	Energy (MWh)	Duration (Hours)	Status	Storage Category
Apple California Flats CA	Apple	2022	60	240	4.0	Operational	Utility-tied (PPA)
Switch Gigawatt 1 Nevada (4 phases)	Switch / Capital Dynamics	2022	200	800	4.0	Operational	On-site BTM (off-grid DC campus)
AES / Microsoft California	Microsoft / AES	2023	55	220	4.0	Operational	Utility DC-PPA
Microsoft Sweden (Saft) — 24 MW / 16 MWh	Microsoft / Saft (TotalEnergies)	2023	24	16	0.7	Operational	On-site BTM (UPS replacement)
AES Baldy Mesa + Silver Peak CA	Amazon / AES	2024	100	400	4.0	Operational	Utility DC-PPA
Eleven Mile Solar Center AZ (Ørsted / Fluence → Meta)	Meta / Ørsted / Fluence	2024	300	1,200	4.0	Operational	Utility DC-PPA
Google BESS Portfolio 2024	Google	2024	250	936	3.7	Operational	Utility DC-PPA
xAI Colossus Phase 1 TN	xAI / Tesla	2024	150	608	4.1	Operational	On-site BTM
xAI Colossus Phase 2 TN	xAI / Tesla	2025	150	655	4.4	Operational	On-site BTM
AES Bellefield 1 CA	Amazon / AES	2025	500	2,000	4.0	Operational	Utility DC-PPA
Google Bell Solar TX (X-ELIO / Fluence)	Google / X-ELIO	2025	300	1,200	4.0	Operational	Utility DC-PPA
Menifee Power Bank CA (Calpine)	Multi-DC Tenants (Calpine)	2025	680	2,720	4.0	Operational	Utility DC-PPA
Greenflash Project Soho TX	Greenflash Infrastructure (H.I.G.)	2026	400	800	2.0	Under Construction	Utility DC-PPA
Iron Mountain NJE-1 NJ	Iron Mountain	2026	12	23	1.9	Under Construction	On-site BTM
Aligned DC + Calibrant OR	Aligned Data Centers / Calibrant Energy	2026	31	62	2.0	Under Construction	On-site BTM
Crusoe / Redwood Energy TX	Crusoe / Redwood Energy	2026	12	63	5.3	Operational	On-site BTM
Platinum Storage TX	Multi-DC Tenants (Engie)	2026	348	1,392	4.0	Operational	Utility DC-PPA
GridStor Oklahoma	GridStor (Goldman Sachs)	2026	200	800	4.0	Under Construction	Utility DC-PPA
Greenflash Rock Rose TX	Greenflash Infrastructure (H.I.G.)	2027	200	400	2.0	Announced	Utility DC-PPA
AES Bellefield 2 CA	Amazon / AES	2027	500	2,000	4.0	Under Construction	Utility DC-PPA
NIPSCO GenCo IN	Amazon / NiSource	2027	400	1,600	4.0	Under Construction	Utility DC-PPA
Cowboy Project WY	Meta / Enbridge	2027	200	1,600	8.0	Announced	Utility DC-PPA
e-Storage / Canadian Solar US	Undisclosed US Utility (DC grid)	2027	500	2,493	5.0	Announced	Utility DC-PPA
DTE / Oracle MI (Stargate) — 8 sites	Oracle / DTE Energy	2028	1,500	6,000	4.0	Announced	Utility DC-PPA

Source: Companies, Bernstein analysis

EXHIBIT 4: **Project-Level Trend of US Data Center BESS Duration and Capacity**

Source: Companies, Bernstein analysis

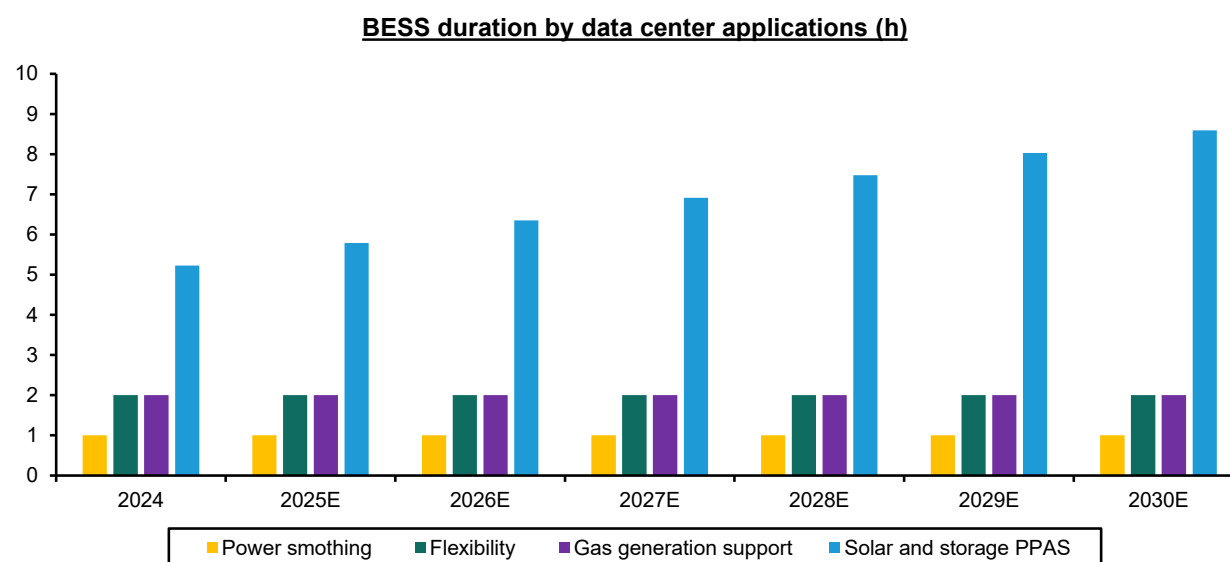
EXHIBIT 5: Behind the meter: Only 5% of the installed base, but 24% of what's under construction, and 39% of the project pipeline



Standed capacity = delayed+cancelled+notapproved/withdrawn capacity
Source: Aterio, Bernstein analysis

Duration requirements differ meaningfully by data center battery application. Based on BNEF data, power smoothing, flexibility, and gas generation support remain relatively stable at around 1 hour, 2 hours, and 2 hours, respectively. However, solar and storage PPAs are expected to require longer-duration BESS solutions, driving the overall duration increase over time. Based on our estimates, solar and storage PPA duration rises from approximately 5 hours in 2024 to over 8 hours by 2030, after adjusting for an attachment rate of 0.61. This suggests that solar-linked data center power procurement could become a key driver of longer-duration battery demand.

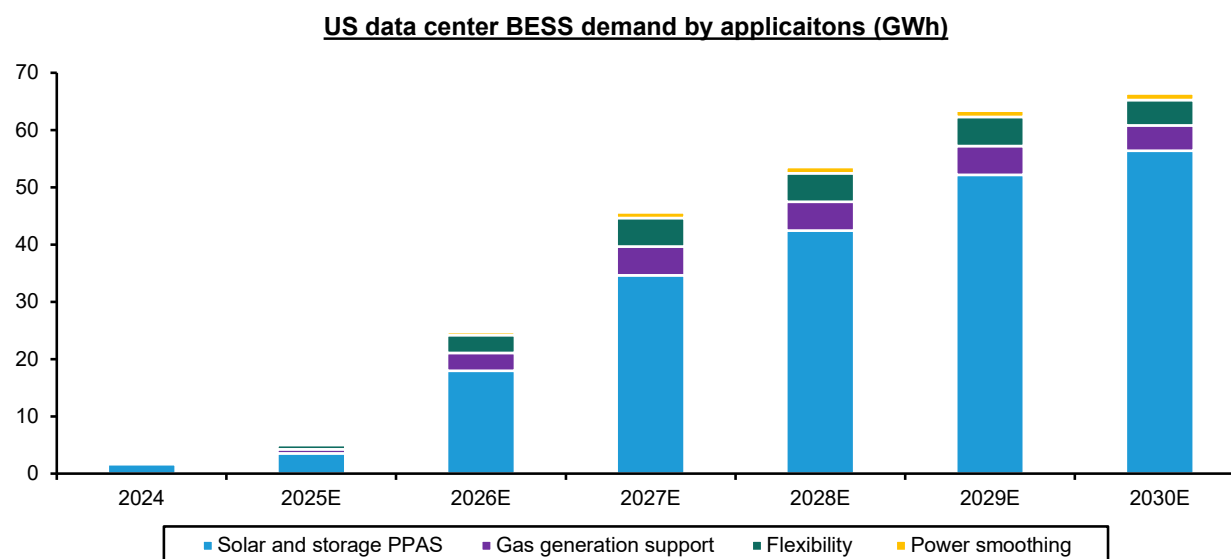
EXHIBIT 6: BESS Duration by Data Center Application (h): Solar and storage PPAs are estimated to reach 8 hours by 2030.



Solar duration is estimated by adjusting for the attachment rate effect, using a factor of 0.61. Source: BNEF.
Source: BNEF, Bernstein analysis and estimation

Based on our estimates, total U.S. data center BESS demand is expected to grow rapidly over the decade, reaching approximately 66 GWh by 2030. Solar and storage PPAs remain the key driver of this growth, reflecting data centers' increasing need for renewable-linked, longer-duration energy storage solutions.

EXHIBIT 7: US data center battery installation by applications



Source: BNEF, Bernstein analysis and estimation

Based on our data center power capacity growth assumptions, we estimate battery demand across three applications: BESS, UPS and BBU. For BESS, we assume penetration rises from 10% in 2024 to 100% by 2030, reflecting increasing adoption of energy storage for data center backup power, peak load management and renewable-linked procurement. We also assume effective BESS duration expands from 4.0 hours in 2024 to 6.0 hours in 2030, calculated as the weighted average duration across different application scenarios. For UPS and BBU, duration requirements remain relatively stable, and we assume 0.15 hours for UPS and 0.08 hours for BBU over the forecast period. On this basis, we estimate cumulative U.S. data center battery installations will reach 277 GWh by 2030, with annual additions of 69 GWh in 2030.

EXHIBIT 8: US Center Battery Demand Forecast and we estimate 69GWh addition in 2030

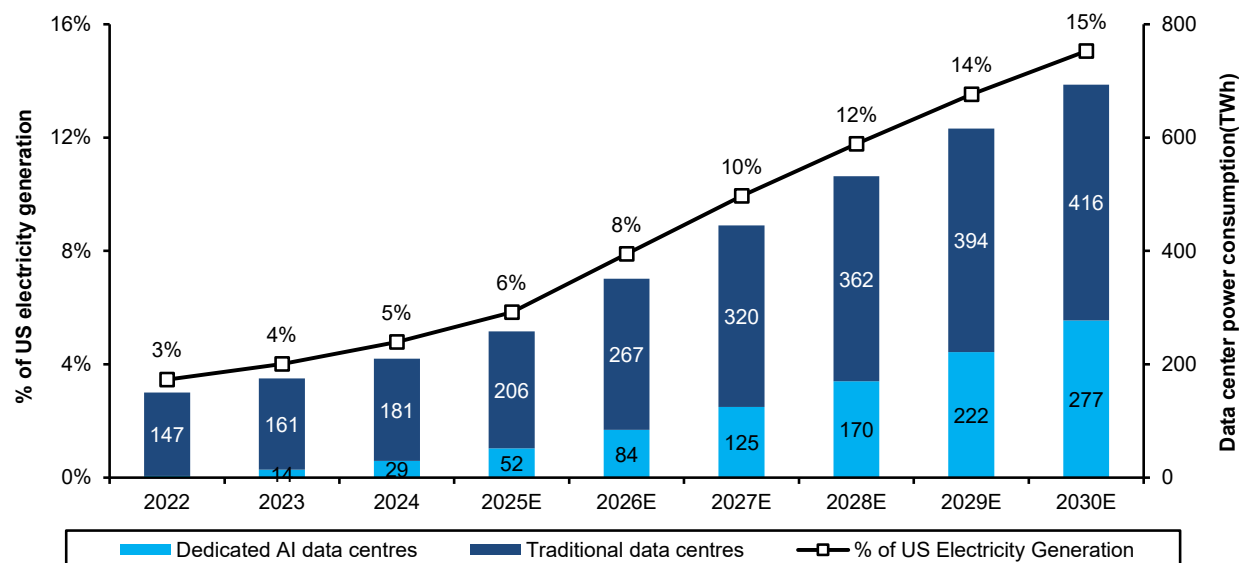
		2020	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
US data center power consumption	TWh	100	118	139	173	218	258	351	445	532	616	694
US data center power capacity	GW	16	19	23	28	35	41	54	68	80	92	103
US data center power (incremental)	GW	2	3	4	5	6	6	14	14	12	12	11
BESS battery penetration	%					10%	20%	40%	70%	85%	95%	100%
Effective capacity	GW					1	1	6	10	10	11	11
BESS duration	hrs					4.0	4.2	4.5	4.8	5.1	5.5	6.0
UPS duration	hrs					0.15	0.15	0.15	0.15	0.15	0.15	0.15
BBU duration	hrs					0.08	0.08	0.08	0.08	0.08	0.08	0.08
Total duration	hrs					4.2	4.5	4.7	5.0	5.4	5.8	6.2
Data center BESS requirement	GWh					3	5	25	46	54	63	66
Data center UPS	GWh					1	1	2	2	2	2	2
Data center BBU	GWh					1	1	1	1	1	1	1
Data center total required (incremental)	GWh					4	7	27	48	56	66	69
Data center total cumulative installation	GWh					4	11	38	86	142	208	277
Multiple						0.1	0.3	0.7	1.3	1.8	2.3	2.7

Source: BNEF, IEA, Bernstein analysis and estimates

U.S. data centers consumed approximately 200 TWh of electricity in 2024, and we expect this to nearly triple by 2030, reaching around 15% of total U.S. power demand. Dedicated AI data centers are expected to become a key growth driver for power demand over the coming years, with electricity consumption rising from 29 TWh in 2024 to 277 TWh by 2030, accounting for

around 40% of total data center power demand.

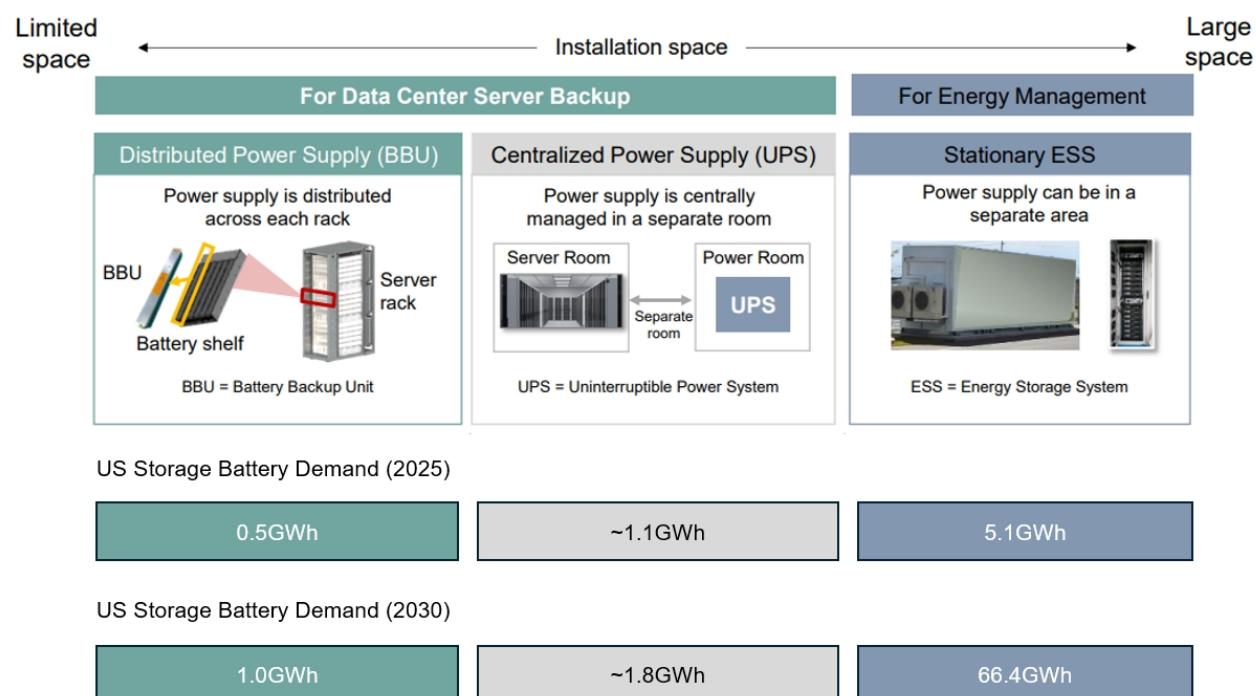
EXHIBIT 9: US data center consumed about 200TWh in 2024 and we expected 15% of electricity consumption in 2030



Source: BNEF, IEA, Bernstein analysis and estimates(2025+)

Among the three main data center power and backup configurations, ESS is expected to be the primary driver of battery demand growth. We estimate U.S. BBU battery demand will increase from 0.5 GWh in 2025 to 1.0 GWh in 2030, while UPS demand will rise from 1.1 GWh to 1.8 GWh. By contrast, ESS demand is expected to grow significantly from 5.1 GWh in 2025 to 66.4 GWh in 2030.

EXHIBIT 10: Power and backup configuration of data centers. BESS used for energy management could be the key driver of battery demand



Source: BNEF, IEA, Panasonic Energy, Synergy Research, Bernstein analysis and estimates

US BATTERY CAPACITY AND DEMAND

Based on our updated assumptions, U.S. battery demand is now expected to grow at a CAGR of approximately 22% through 2030, supported by EV demand but increasingly driven by ESS and data center demand. However, demand growth is still expected to lag the pace of new manufacturing capacity additions. U.S. battery manufacturing capacity is projected to expand from 167 GWh at year-end 2025 to 294 GWh by year-end 2026, and could reach around 721 GWh by 2030, indicating that structural overcapacity is likely to persist through the decade.

So far in 2026, several companies have announced changes to their capacity plans. Some are shifting capacity from EV batteries to ESS, some are moving from full utilization to partial utilization, while others have pursued sale-and-leaseback arrangements to improve financial performance and reduce costs. However, most announcements provide limited detail on the scale and timing of capacity transfers or adjustments. Based on the announced changes so far, our 2030 U.S. capacity estimate has been revised down from 798 GWh to 721 GWh. We expect further announcements could help bring U.S. capacity closer to a more reasonable level.

At the same time, stronger ESS and data center demand leads us to raise our utilization rate assumptions. Our 2030 utilization rate estimate is now 15% higher than our previous forecast. We also raise our total U.S. battery demand estimate to 486 GWh in 2030, which is 74 GWh higher than our previous estimate. By segment, EV battery demand is revised lower, but this is more than offset by sharply higher demand from ESS and data centers. Driven by stronger demand growth from ESS and data centers, we estimate that ESS will account for 40% of total battery demand by 2030, with data centers contributing a further 14%.

As a result, we expect U.S. battery overcapacity to persist for most of the decade, with utilization rates remaining under pressure. This could increase the likelihood of project delays, phased commissioning schedules, or outright cancellations as producers reassess capex plans. In this context, we believe investors should focus on low-cost, well-contracted players that are better positioned to sustain cash generation in an environment where supply continues to outpace demand growth.

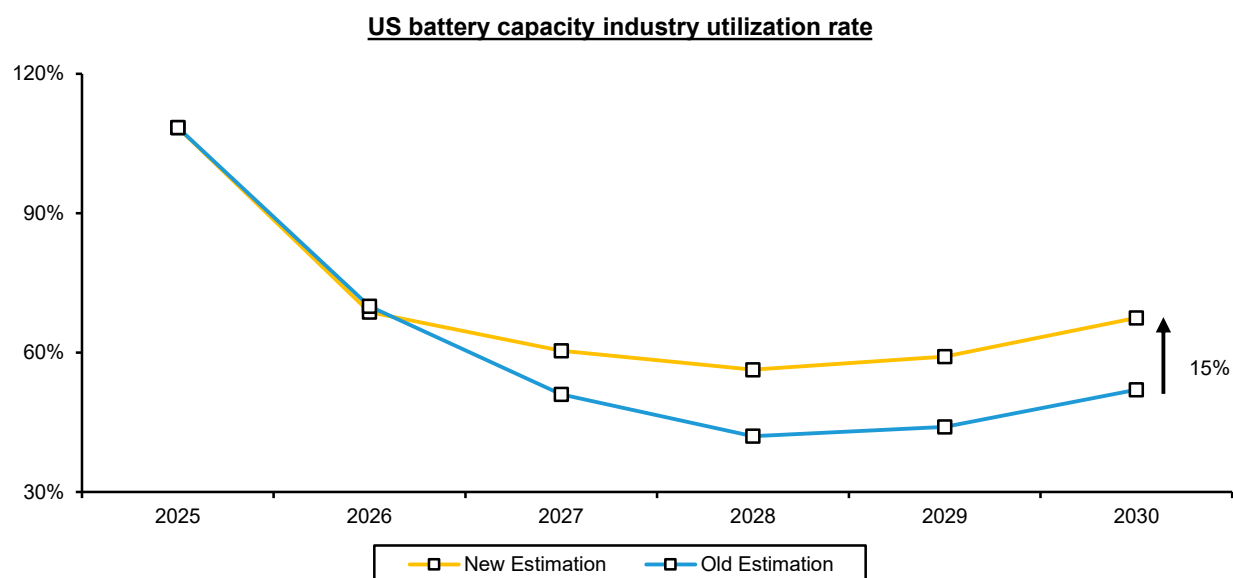
EXHIBIT 11: **U.S. Battery Capacity vs. Demand Outlook: Data Centers to Account for ~15% of Total Demand by 2030**

US Market	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2025-30 CAGR
US Demand														
EV	20	20	22	43	71	109	130	134	109	134	153	179	220	10%
ESS	1	1	3	11	12	21	34	42	66	108	166	175	197	36%
Data center							4	7	27	48	56	66	69	59%
Total Demand	21	21	25	54	83	131	169	182	202	289	375	421	486	22%
Demand % YoY		-1%	19%	118%	53%	57%	29%	8%	11%	43%	29%	12%	16%	
ESS % of Demand	4%	4%	13%	20%	15%	16%	20%	23%	33%	37%	44%	42%	40%	
Data center % of Demand							2%	4%	13%	17%	15%	16%	14%	
Total Available Capacity														
LGES	5	5	5	5	5	15	50	75	130	197	232	232	232	25%
SDI	0	0	0	0	0	0	0	0	15	42	64	64	73	37%
Tesla	0	0	0	0	10	10	10	28	28	38	83	128	128	36%
SK On	0	0	0	0	10	10	22	39	39	79	102	102	102	36%
Panasonic	23	23	31	31	31	38	38	40	57	74	75	76	77	14%
Others	3	3	3	3	3	3	3	3	25	50	110	110	110	106%
Total Available Capacity	31	31	39	39	59	76	123	167	294	479	665	711	721	34%
Demand % Capacity	68%	68%	64%	139%	141%	172%	138%	109%	69%	60%	56%	59%	67%	

SDI, LGES were covered, all others not

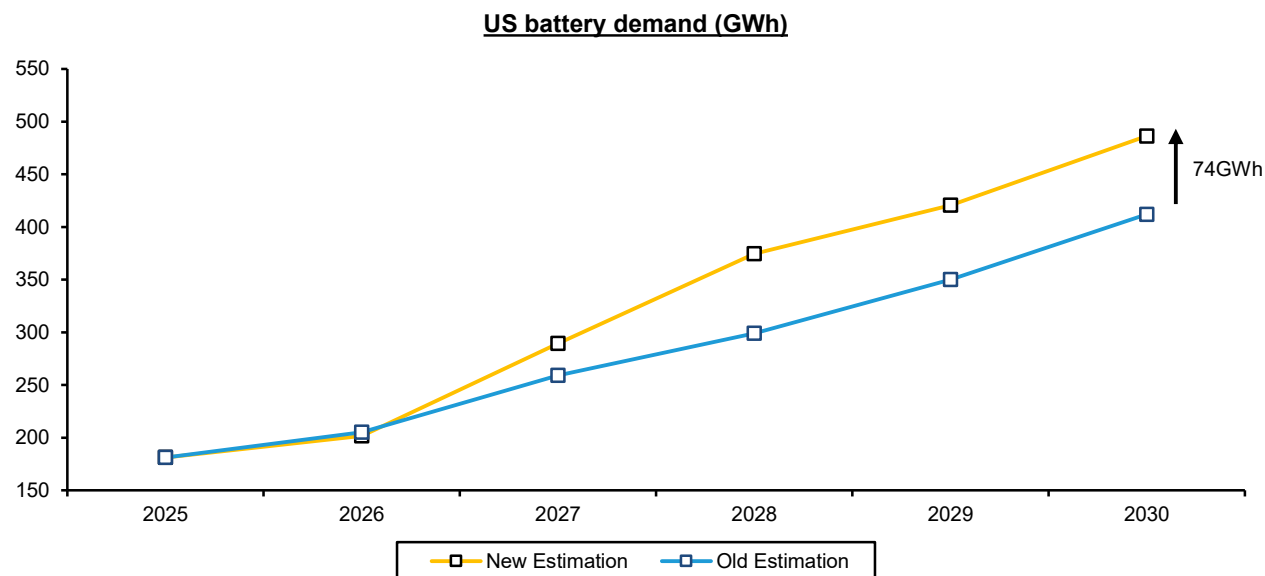
Source: Company data, Bernstein estimates and analysis

Driven by strong demand growth from ESS and data centers, as well as reduced capacity in the U.S., we estimate that the U.S. industry utilization rate in 2030 will be 15% higher than our previous forecast.

EXHIBIT 12: **We have raised our 2030 utilization rate estimate by 15% compared with our previous forecast.**

Source: Company data, Bernstein estimates and analysis

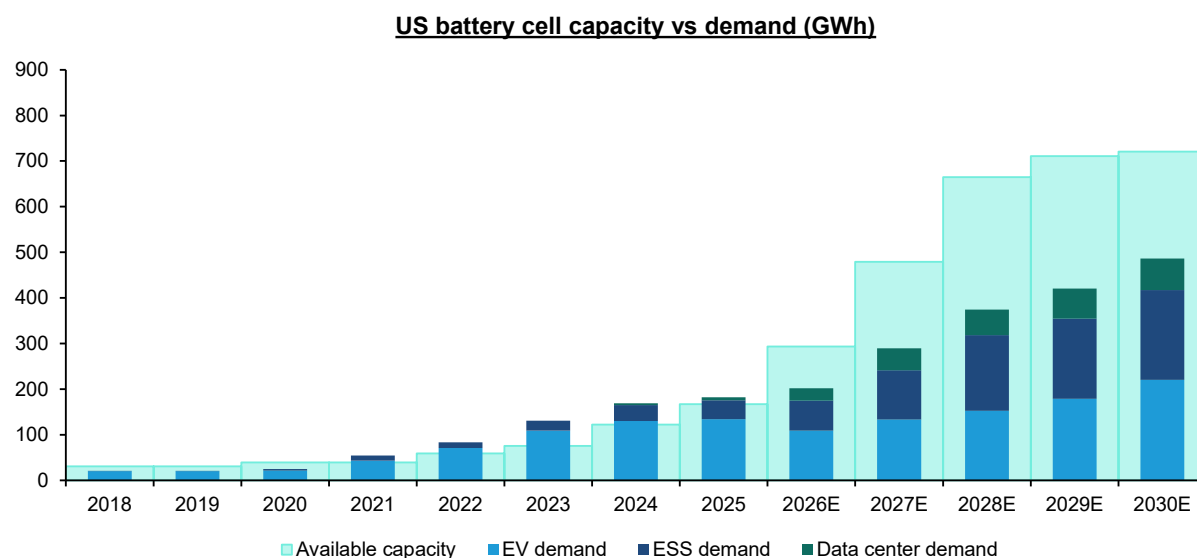
We also revise up our U.S. battery demand estimate by 74 GWh by 2030 compared with our previous forecast.

EXHIBIT 13: **We raise up our near term demand forecast and in 2030 raised 74GWh compare to last estimation**

Source: Company data, Bernstein estimates and analysis

ESS and data center demand are accelerating, while EV demand is slowing. ESS demand continues to grow rapidly, supported by rising power needs, higher renewable penetration, and policy incentives such as the ITC and AMPC. Utilities are adding storage to firm solar and wind generation, replace aging gas peakers, and improve grid reliability, particularly in markets such as California and Texas. At the same time, AI and hyperscale data centers are emerging as an important new source of battery demand, driven by backup power and peak load management needs. We estimate data centers will account for 14% of total U.S. battery demand by 2030, making this segment a key driver of incremental growth. By contrast, EV battery demand has weakened as mass-market adoption slows, subsidies are rolled back, and OEMs shift production priorities toward hybrids and ICE vehicles. Reflecting stronger ESS and data center demand, we raise our 2030 U.S. battery demand estimate by 74 GWh versus our previous forecast, with ESS and data centers expected to represent 40% and 14% of total demand, respectively, by 2030. Despite this demand uplift, supply growth remains faster than demand growth. We revise our 2030 U.S. battery capacity estimate down from 798 GWh to 721 GWh based on announced capacity adjustments, but this still exceeds our 2030 demand estimate of 486 GWh. As a result, we expect structural oversupply to persist through most of the decade. However, utilization rates could improve further if additional capacity pauses, cancellations, or delayed ramp-ups are announced.

EXHIBIT 14: **Despite strong growth in ESS and data center demand, we expect structural overcapacity in the U.S. to persist unless there are further announcements to reduce total capacity.**



Source: Company data, Bernstein estimates and analysis

Company-level capacity details are shown below. LGES remains the largest player, although we revise its capacity estimate down to 232 GWh following the pause of its Ohio battery plant. In their 1Q results conference meetings, both SDI and LGES management indicated plans to convert some EV battery production lines to ESS. However, the detailed scale and timing of these capacity adjustments have not yet been announced.

EXHIBIT 15: **US battery production capacity(year end)**

YE US Battery Capacity (GWh)	2024	2025	2026E	2027E	2030E
LGES	75	130	197	232	232
SDI	0	15	42	64	73
Tesla	28	28	38	83	128
SK On	22	39	79	102	102
Panasonic	40	57	74	75	78
Ford	0	0	10	40	57
Others	3	25	40	70	63
Total	167	294	479	665	732

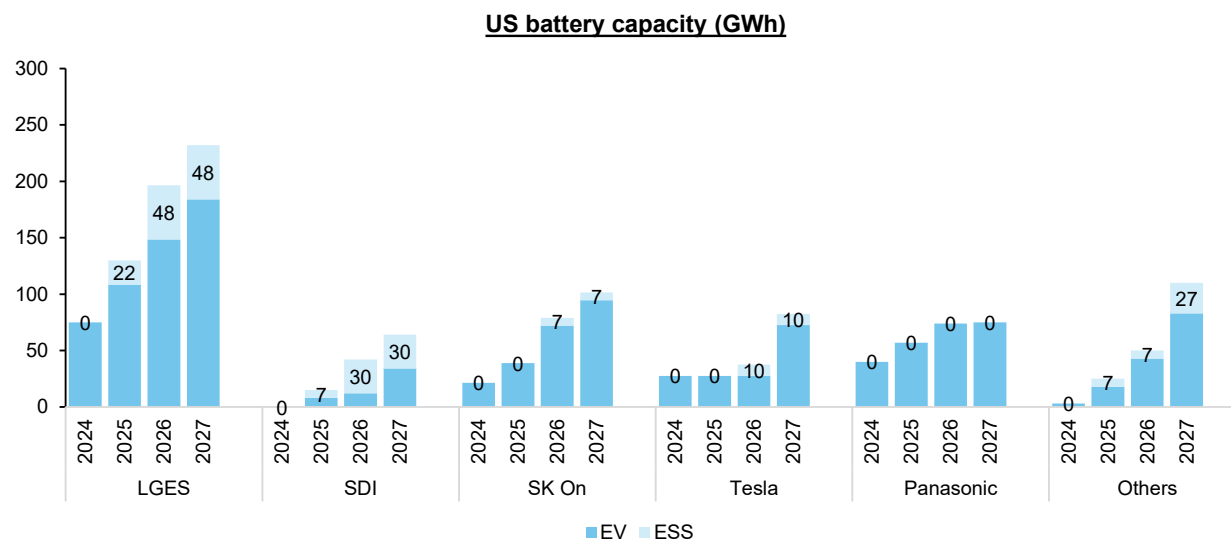
YE US ESS Capacity (GWh)	2024	2025	2026E	2027E
LGES	0	22	48	48
SDI		7	30	30
Tesla			10	10
SK On			7	7
Panasonic			0	0
Ford			0	20
Others		7	7	7
Total	0	36	102	122

% ESS	2024	2025E	2026E	2027E
LGES	0%	17%	24%	21%
SDI		47%	71%	47%
Tesla	0%	0%	27%	12%
SK On	0%	0%	9%	7%
Panasonic	0%	0%	0%	0%
Ford			0%	50%
Others	0%	28%	18%	10%
Total	0%	12%	21%	18%

LGES, Samsung SDI are covered by Neil Beveridge. All other stocks are not covered by Bernstein

Source: Company data, Bernstein estimates and analysis

ESS cell production in the U.S. is likely to remain a bottleneck in 2026, as localized ESS supply may still lag demand from grid-scale storage projects and AI data centers. Based on our updated capacity assumptions, U.S. localized ESS battery capacity could reach around 102 GWh by year-end 2026 and increase further to around 122 GWh by 2027, supported by LGES, Samsung SDI, SK On, Tesla and other manufacturers shifting capacity toward ESS and LFP applications. LGES remains the largest U.S. battery producer, with total capacity expected to reach around 198 GWh by year-end 2026 and 232 GWh by year-end 2027. Its dedicated ESS capacity is expected to reach 48 GWh from 2026 onward. Samsung SDI appears to be making the most aggressive near-term transition to ESS, with ESS capacity expected to reach 30 GWh by 2026, representing more than half of its U.S. battery manufacturing capacity. SK On and Tesla are also expected to add ESS capacity from 2026, while other manufacturers could become a more meaningful contributor by 2027. With ESS and data center demand accelerating — and data centers expected to account for 14% of total U.S. battery demand by 2030 — localized ESS supply will likely become increasingly important. While more than 100 GWh of domestic ESS capacity by end-2026 could help meet a meaningful portion of U.S. demand and reduce reliance on China's ESS battery supply chain, ramp-up timing and execution remain key risks. Further announcements of EV-to-ESS capacity shifts could help ease supply constraints and improve the domestic ESS supply-demand balance

EXHIBIT 16: **US battery capacity by key suppliers**

SDI, LGES were covered, All others not

Source: Company data, Bernstein estimates and analysis(2026+)

COMPANY OUTLOOK**Model update :**

SDI: We maintain our TP unchanged. However, we revise our EPS estimates, lowering FY26 EPS from 2,099 to 1,752, while slightly increasing FY27 EPS from 18,376 to 18,577. Based on the demand update we adjust the battery sales volume, but we did not change any other multiples and assumptions

LGES: We modestly raise our TP from KRW347,000 to KRW351,000 while maintaining our rating. We lower our FY26 EPS estimate from 1,811 to -2,364, and raise our FY27 EPS estimate from 9,453 to 10,255. Based on the demand update we adjust the battery sales volume, but we did not change any other multiples and assumptions

LG Chem: We modestly increase our TP from KRW298,000 to KRW312,000 while maintaining our rating. We lower our FY26 EPS estimate from 3,030.56 to 2,693, and increase our FY27 EPS estimate from 27,968 to 28,743. Based on the demand update we adjust the battery sales volume, but we did not change any other multiples and assumptions

SAMSUNG SDI (MARKET-PERFORM, TP KRW520,000/SH)**EXHIBIT 17: Overview of Samsung SDI's battery operation**

Samsung SDI													2025-30
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	CAGR
EV	10	14	24	41	60	54	46	46	55	70	93	119	21%
%YoY		40%	73%	69%	46%	-10%	-15%	0%	20%	28%	32%	28%	
ESS(including data center)	4	4	4	4	4	10	13	27	58	87	120	155	64%
%YoY		0%	-7%	13%	0%	138%	30%	105%	117%	50%	39%	29%	
Total battery shipments (GWh)	14	18	28	45	64	64	59	73	113	157	213	274	36%
%YoY		29%	56%	62%	42%	0%	-8%	23%	55%	39%	36%	29%	
ESS % of Total shipment	29%	22%	13%	9%	7%	16%	22%	37%	51%	55%	56%	57%	
China	4	7	7	7	14	14	14	14	14	14	14	14	0%
US	0	0	0	0	0	0	15	42	64	64	73	73	37%
Europe	8	17	24	31	37	37	47	47	57	67	77	87	13%
Other	6	6	6	12	12	12	38	38	38	38	38	38	0%
Total battery capacity(GWh)	18	30	37	49	63	63	114	141	173	183	202	212	13%
%YoY		67%	23%	34%	27%	0%	81%	24%	23%	6%	10%	5%	
US % of Total capacity	0%	0%	0%	0%	0%	0%	13%	30%	37%	35%	36%	34%	
Utilization	78%	60%	76%	92%	102%	102%	52%	51%	65%	86%	106%	129%	
OPM (excluding AMPC)	0%	3%	5%	7%	7%	1%	-16%	-3%	3%	6%	6%	6%	
OPM (including AMPC)	2%	3%	5%	7%	7%	2%	-12%	1%	10%	14%	15%	12%	

Source: Company data, SNE, Bernstein estimates and analysis(2026+estimate)

EXHIBIT 18: We expect Samsung SDI will achieve a 4% CAGR over the period 2030-2050, while sustaining a market share of 3.4%

Samsung SDI M/S (in GWh)		2023	2024	2025	2035	2040	2050
2030-50 cagr	3.0%	7.2%	5.3%	3.4%	2.7%	2.7%	2.8%
	4.0%	7.2%	5.3%	3.4%	2.7%	3.0%	3.4%
	5.0%	7.2%	5.3%	3.4%	2.7%	3.3%	4.1%
	6.0%	7.2%	5.3%	3.4%	2.7%	3.7%	5.0%
	7.0%	7.2%	5.3%	3.4%	2.7%	4.0%	6.0%

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

EXHIBIT 19: SDI target price sensitivity to revenue cagr and EBIT margin

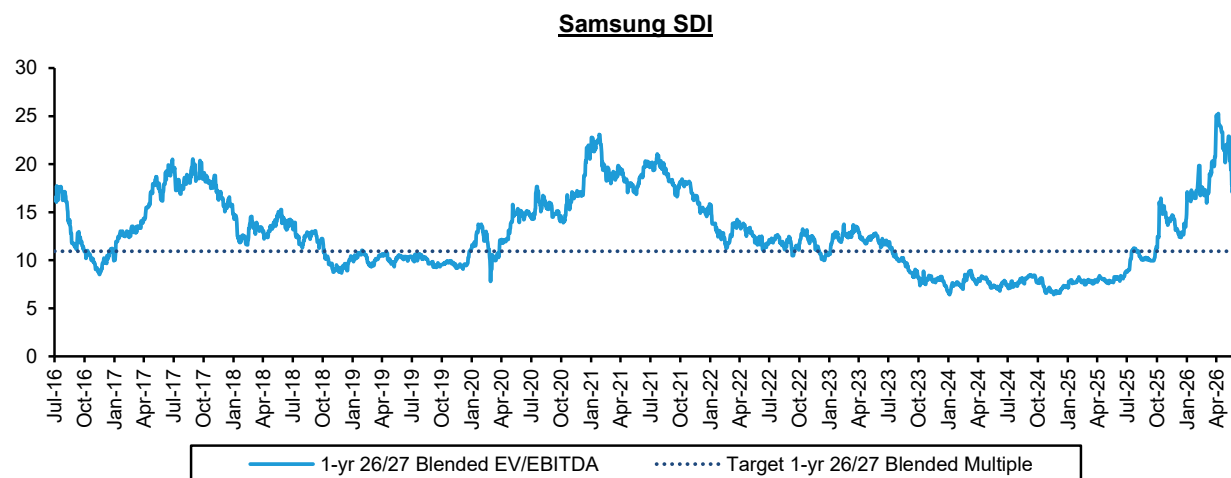
Samsung SDI TP (K KRW)		EBIT margin				
		7.0%	8.0%	9.0%	10.0%	11.0%
2030-50 cagr	3.0%	380	440	500	550	610
	4.0%	430	500	560	620	690
	5.0%	490	560	630	700	780
	6.0%	560	640	720	800	880
	7.0%	630	730	820	910	1,000

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

EXHIBIT 20: **Samsung SDI DCF valuation**

Samsung SDI - Discounted Cash Flow												
KRW bn	2022	2023	2024	2025	2026	2027	2028	2029	2030	2025-2030	2031-2050E	Terminal
Revenue	20,124	22,708	16,592	13,267	16,072	22,041	28,333	35,757	43,352			
Sales Growth	48.5%	12.8%	(26.9%)	(20.0%)	21.1%	37.1%	28.5%	26.2%	21.2%	26.7%	4.0%	3.0%
EBITDA	3,272	3,365	2,262	374	2,064	3,253	4,414	5,110	5,714			
EBIT (ex-AMPC)	1,808	1,637	387	(1,673)	(262)	834	1,910	2,520	3,038			
EBIT Margin	9.0%	7.2%	2.3%	(12.6%)	(1.6%)	3.8%	6.7%	7.0%	7.0%	4.0%	8.4%	
Tax	613	420	7	(495)	(7)	210	481	634	764			
% of EBIT	33.9%	25.7%	1.7%	29.6%	2.6%	25.2%	25.2%	25.2%	25.2%	24.9%	25.2%	
NOPAT	1,195	1,217	381	(1,178)	(255)	624	1,430	1,886	2,274			
D&A	1,463	1,728	1,875	2,047	2,326	2,418	2,504	2,590	2,676			
% of sales	7.3%	7.6%	11.3%	15.4%	14.5%	11.0%	8.8%	7.2%	6.2%	9.2%	8.2%	
Change in NWC:	(1,013)	961	(2,466)	1,585	-	-	-	-	-			
% of sales	(5.0%)	4.2%	(14.9%)	11.9%	0.0%	0.0%	0.0%	0.0%	0.0%	1.0%	0.0%	
(CapEx)	(2,518)	(4,254)	(6,600)	(3,330)	(3,163)	(3,163)	(3,243)	(3,324)	(3,407)			
% of sales	12.5%	18.7%	39.8%	25.1%	19.7%	14.4%	11.4%	9.3%	7.9%	12.4%	8.2%	
Unlevered FCF	(872)	(348)	(6,810)	(876)	(1,093)	(121)	691	1,152	1,543	1,297	84,402	98,035
Our TP implied EV/EBITDA	11.2x	10.9x	15.7x	56.6x	14.0x	7.9x	5.6x	4.6x	4.6x			
Valuation				WACC				Growth 2025-2030				
WACC	9.3%			Risk Free Rate	2.5%			Reinvestment Rate	73%			
Assumed Perpetual Growth	3.0%			Levered Equity Beta	2.0			ROE	7%			
Enterprise Value (EV)	36,686			Market Risk Premium	5.7%			Implied Growth Rate	5%			
- Debt	(11,236)			Cost of Equity	13.6%			NOPAT CAGR	35%			
- Minority, Preferred and Others	(2,183)			After-tax Cost of Debt	3.1%			2030 Capex/DD&A	1.3x			
+ Cash	1,981			Debt / (Market Cap + Debt)	41.3%							
+ Investment	5,846			NPV				NPV (%)				
+ IRA credit	3,804			2025-2030	1,341			2025-2030	4%			
Equity Value	34,897			2031-2050	23,677			2031-2050	65%			
#shares	67			Terminal	11,667			Terminal	32%			
DCF result (KRW)	520,000			Total	36,686			Total	100%			

Source: Company data, Bernstein estimates and analysis

EXHIBIT 21: **Our TP implies a 1-yr 26/27 blended EV/EBITDA of 11x for SDI**

Source: Bloomberg, Bernstein analysis

LGES (MARKET-PERFORM, TP KRW351,000/SH)**EXHIBIT 22: Overview of LGES's battery operation**

LGES													
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2025-30 CAGR
EV	27	48	85	101	136	137	152	168	205	253	322	403	21%
%YoY		80%	77%	19%	35%	1%	11%	11%	22%	23%	27%	25%	
ESS(including data center)	3	6	8	9	8	8	15	42	88	127	163	200	68%
%YoY		109%	38%	13%	-11%	0%	88%	180%	109%	45%	29%	23%	
Total battery shipments (GWh)	29	54	93	110	144	146	167	211	293	380	485	603	29%
%YoY		85%	72%	18%	31%	1%	15%	26%	39%	30%	28%	24%	
ESS % of Total shipment	9%	11%	9%	8%	6%	5%	9%	20%	30%	33%	34%	33%	
China	27	27	27	93	123	123	153	153	153	153	153	153	0%
US	5	5	5	15	50	75	130	197	232	232	232	232	12%
Europe	40	70	70	70	90	100	115	115	115	115	115	115	0%
Other	6	6	14	21	21	56	81	94	94	94	94	94	3%
Total battery capacity(GWh)	78	108	116	199	284	354	479	558	594	594	594	594	4%
% YoY		38%	7%	72%	43%	24%	35%	17%	6%	0%	0%	0%	
US % of Total capacity	6%	5%	4%	8%	18%	21%	27%	35%	39%	39%	39%	39%	
Utilization	38%	50%	81%	55%	51%	41%	35%	38%	49%	64%	82%	102%	
OPM (excluding AMPC)	-5%	-1%	4%	5%	4%	-4%	-1%	-3%	3%	6%	7%	10%	
OPM (including AMPC)	-5%	-1%	4%	5%	6%	2%	6%	4%	13%	16%	18%	20%	

Source: Company data, SNE, Bernstein estimates and analysis (2026+ estimates)

EXHIBIT 23: We expect LGES will achieve a 7% CAGR over the period 2030-2050, and its market share will increase from current 10% to 13% by 2050

LGES M/S (in GWh)		2023	2024	2025	2035	2040	2050
2030-50 cagr	1.0%	16.2%	12.1%	9.6%	6.0%	5.1%	4.2%
	3.0%	16.2%	12.1%	9.6%	6.0%	6.2%	6.3%
	5.0%	16.2%	12.1%	9.6%	6.0%	7.5%	9.2%
	7.0%	16.2%	12.1%	9.6%	6.0%	9.0%	13.3%
	9.0%	16.2%	12.1%	9.6%	6.0%	10.8%	19.5%

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

EXHIBIT 24: LGES target price sensitivity to revenue cagr and EBIT margin

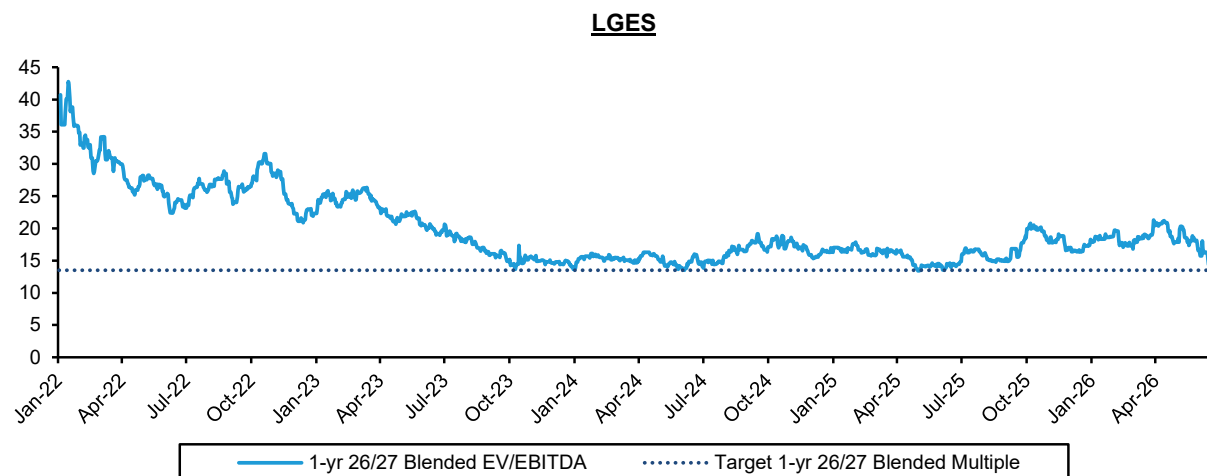
LGES TP (K KRW)		EBIT margin				
		7.0%	8.0%	9.0%	10.0%	11.0%
2030-50 cagr	1.0%	107	128	150	172	194
	3.0%	144	171	199	226	253
	5.0%	195	229	264	298	332
	7.0%	263	307	351	395	439
	9.0%	354	411	468	525	582

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

EXHIBIT 25: **LGES DCF valuation**

LGES - Discounted Cash Flow												
KRW bn	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2025-2030	2031-2050E	Terminal
Revenue	25,599	33,746	25,620	23,671	30,769	41,388	52,227	64,948	78,844			
Sales Growth	43.4%	31.8%	(24.1%)	(7.6%)	30.0%	34.5%	26.2%	24.4%	21.4%	27.2%	7.0%	3.0%
EBITDA	3,046	3,770	2,137	3,391	3,268	5,597	7,396	8,832	12,449			
EBIT (ex-AMPC)	1,214	1,486	(906)	(302)	(1,044)	1,231	2,957	4,327	7,884			
EBIT Margin	4.7%	4.4%	(3.5%)	(1.3%)	(3.4%)	3.0%	5.7%	6.7%	10.0%	5.2%	9.0%	
Tax	215	407	11	333	453	308	739	1,082	1,971			
% of EBIT	17.7%	27.4%	(1.2%)	(110.3%)	(43.4%)	25.0%	25.0%	25.0%	25.0%	32.5%	25.0%	
NOPAT	999	1,080	(917)	(635)	(1,497)	923	2,218	3,245	5,913			
D&A	1,832	2,284	3,043	3,693	4,312	4,366	4,439	4,505	4,565			
% of sales	7.2%	6.8%	11.9%	15.6%	14.0%	10.5%	8.5%	6.9%	5.8%	8.9%	8.0%	
Change in NWC:	(7,949)	1,364	1,811	402	-	-	-	-	-			
% of sales	(31.1%)	4.0%	7.1%	1.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	
(CapEx)	(6,291)	(10,890)	(12,548)	(10,417)	(5,727)	(5,155)	(5,152)	(5,377)	(5,380)			
% of sales	24.6%	32.3%	49.0%	44.0%	18.6%	12.5%	9.9%	8.3%	6.8%	12.7%	8.0%	
Unlevered FCF	(11,409)	(6,162)	(8,611)	(6,957)	(2,913)	134	1,505	2,373	5,098	(760)	233,448	319,661
AMPC	-	646	1,566	1,526	2,379	4,144	5,595	7,553	8,081			
Our TP implied EV/EBITDA	31.7x	21.7x	26.7x	19.2x	17.1x	9.9x	7.4x	5.9x	4.7x			
Valuation			WACC				Growth 2025-2030					
WACC	9.6%					2.5%						107%
Assumed Perpetual Growth	3.0%					1.5						10%
Enterprise Value (EV)	96,555					5.7%						11%
						10.8%						n.a.
- Debt	(24,682)					3.3%						
- Minority, Preferred and Others	(10,402)					15.2%						1.2x
+ Cash	2,930											
+ IRA credit	17,671											
Equity Value	82,073											
			NPV				NPV (%)					
						3,791						4%
						57,622						60%
#shares	234					35,142						36%
DCF result (KRW)	351,000					96,555						100%

Source: Company data, Bernstein estimates and analysis

EXHIBIT 26: **Our TP implies a 1-yr 26/27 blended EV/EBITDA of 14x for LGES**

Source: Bloomberg, Bernstein analysis

LG Chem (Market-perform, TP KRW312,000/sh)

EXHIBIT 27: **LG Chem SOTP model**

LG Chem SOTP Segment	Avg '25-28 NP KRW bn	Multiple	Valuation Method	USD bn	KRW bn	KRW/share
Petrochemicals	197	10.0	P/E ('25-27)	1.4	1,967	25,000
Advanced materials	238	2.0	EV/S (2025)	5.7	8,175	104,000
Life sciences	67	50.0	P/E ('25-27)	2.4	3,370	43,000
Energy solutions (79.38% interest)	1,199		DCF	63.8	90,673	1,158,000
Cash and cash eqv.				7.7	10,903	139,000
Debt				-23.8	-33,812	-432,000
Less: Net Debt (Cash)				16.1	22,909	293,000
Total Value				57.2	81,276	1,038,000
Conglomerate Discount (70%)						312,000

Source: Company data, Bernstein estimates and analysis

EXHIBIT 28: LGES financial exhibit

Income Statement (KRW bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Revenue	12,570	17,852	25,599	33,746	25,620	23,671	30,769	41,388	52,227
COGS	(10,568)	(13,953)	(21,308)	(28,802)	(22,214)	(19,440)	(26,058)	(33,121)	(40,391)
Gross Profit	2,002	3,899	4,291	4,944	3,406	4,231	4,711	8,267	11,836
SG&A	(2,292)	(3,130)	(3,077)	(3,456)	(4,311)	(4,533)	(3,376)	(2,892)	(3,284)
Operating Profit	(290)	768	1,214	2,163	574	1,346	1,335	5,375	8,552
Finance income/(expense)	133	340	(133)	341	629	1,103	1,585	1,644	1,416
Investment Income	(1)	(12)	(37)	(35)	(34)	(34)	(34)	(34)	(34)
Other non-operating income/(expense)	648	465	(48)	(425)	(820)	(2,000)	(1,904)	(1,200)	(1,200)
Income Before Taxes	(701)	777	996	2,044	349	415	982	5,786	8,735
Income Tax Expense (Income)	(124)	77	215	407	11	333	453	1,157	1,747
Profit (loss) from discontinued operations	127	229	-	-	-	-	-	-	-
Net Income	(449)	930	781	1,638	338	82	529	4,629	6,988
Minority Interest/Preferred	-	185	(5)	401	1,357	1,154	1,082	2,236	2,365
Net Income to Common	(449)	745	786	1,237	(1,019)	(1,072)	(553)	2,393	4,622
EBITDA	776	2,220	3,046	4,447	3,617	5,039	5,647	9,741	12,991
EPS (KRW)	(2,247)	3,963	3,306	5,287	(4,354)	(4,581)	(2,364)	10,225	19,754
BPS (KRW)	37,827	57,045	86,751	104,162	132,338	125,308	134,140	153,921	183,784
Balance Sheet (KRW bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & Cash Equivalents	1,493	1,283	5,938	5,069	3,899	3,779	2,930	4,647	8,605
Accounts Receivables	2,931	2,914	4,772	5,129	4,944	4,311	7,282	9,796	12,361
Inventories	3,043	3,896	6,996	5,396	4,552	4,350	6,117	7,641	9,518
Other Current Assets	1,618	1,443	1,443	1,443	1,443	1,443	1,443	1,443	1,443
Current Assets	9,086	9,536	19,149	17,037	14,838	13,883	17,772	23,526	31,927
Property, plant & equipment	8,684	11,051	15,331	23,655	38,350	40,795	43,367	44,126	44,812
Intangible Assets/Goodwill	326	455	613	946	1,534	1,632	1,735	1,765	1,792
Investments	48	226	226	226	226	226	226	226	226
Other Non-current Assets	1,798	2,497	2,981	3,574	5,360	10,613	11,642	11,642	11,642
Non-Current Assets	10,856	14,228	19,150	28,400	45,469	53,265	56,970	57,759	58,473
Total Assets	19,942	23,764	38,299	45,437	60,307	67,148	74,742	81,285	90,399
Accounts Payable	2,592	2,178	3,842	3,094	2,706	2,153	3,670	4,584	5,711
Short-Term Borrowings	1,123	2,203	2,773	4,185	6,414	9,975	11,060	11,060	11,060
Other Short-Term Liabilities	3,174	5,093	5,093	5,093	5,093	5,093	5,093	5,093	5,093
Current Liabilities	6,889	9,474	11,708	12,372	14,213	17,221	19,823	20,737	21,863
Long-Term Borrowings	5,078	4,766	5,336	6,747	8,977	12,537	13,622	13,622	13,622
Other Long-Term Liabilities	409	782	662	1,944	6,150	8,068	9,157	9,157	9,157
Long-term Liabilities	5,487	5,548	5,998	8,691	15,127	20,605	22,779	22,779	22,779
Total Liabilities	12,376	15,022	17,706	21,063	29,340	37,826	42,602	43,516	44,643
Shareholders' Equity	7,565	8,742	20,593	24,374	30,967	29,322	31,389	36,018	43,005
Total Liabilities and Equity	19,942	23,764	38,299	45,437	60,307	67,148	73,991	79,534	87,648
Net Debt (Cash)	4,708	5,686	2,171	5,863	11,492	18,733	21,752	20,035	16,077
Cash Flow Statement (KRW bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Net Income	(449)	930	781	2,092	338	82	529	4,629	6,988
Depreciation & Amortization	1,066	1,452	1,832	2,284	3,043	3,693	4,312	4,366	4,439
Changes in Net Working Capital	1,493	215	-	-	-	-	(2,175)	(3,123)	(3,316)
Other	(1,407)	(1,618)	(3,193)	68	1,730	657	283	1,000	1,000
Cash Flow From Operations	703	979	(580)	4,444	5,111	4,432	2,949	6,872	9,110
Capex	(2,700)	(4,000)	(6,291)	(10,890)	(12,548)	(10,417)	(5,727)	(5,155)	(5,152)
Other	(947)	1,822	29	1,170	482	(465)	628	-	-
Cash Flow From Investing	(3,647)	(2,178)	(6,262)	(9,720)	(12,066)	(10,882)	(5,099)	(5,155)	(5,152)
Borrow/Repay	n.a.	556	810	2,823	3,590	7,122	2,170	-	-
Other	n.a.	327	10,604	1,532	1,792	(837)	(981)	-	-
Cash Flow From Financing	3,164	883	11,414	4,355	5,382	6,285	1,189	-	-
Net Cash Flow	n.a.	(317)	4,572	(912)	(1,573)	(165)	(962)	1,717	3,958
Cash at Beginning of Period	n.a.	7,470	7,259	11,914	11,045	9,875	9,755	8,907	10,623
FX effect	n.a.	106	83	43	403	45	113	-	-
Cash at End of Period	1,493	7,259	11,914	11,045	9,875	9,755	8,907	10,623	14,582
Free Cash Flow	(1,997)	(3,021)	(6,871)	(6,446)	(7,437)	(5,985)	(2,779)	1,717	3,958

Source: Company data, Bernstein estimates and analysis

EXHIBIT 29: Samsung SDI financial exhibit

Income Statement	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Revenue	11,295	13,553	20,124	22,708	16,592	13,267	16,072	22,041	28,333
COGS	8,914	10,476	15,903	18,726	13,498	11,806	13,007	16,646	20,560
Gross Profit	2,381	3,078	4,221	3,982	3,094	1,461	3,065	5,395	7,773
SG&A	1,709	2,010	2,413	2,349	2,731	3,183	2,850	3,155	3,755
Operating Profit	671	1,068	1,808	1,633	363	(1,722)	215	2,240	4,019
Finance income/(expense)	(63)	(63)	(63)	(63)	(63)	(63)	(129)	(150)	1
Investment Income	293	596	844	853	169	359	281	229	234
Other non-operating income/(expense)	(98)	63	63	63	58	62	64	64	64
Income Before Taxes	804	1,663	2,652	2,486	527	(1,364)	430	2,383	4,318
Income Tax Expense (Income)	173	413	613	420	7	(495)	(7)	465	842
Discontinued operations	-	-	-	-	55	284	-	-	-
Net Income	631	1,250	2,039	2,066	576	(585)	437	1,918	3,476
Minority Interest/Preferred	144	82	89	90	(24)	65	322	703	1,054
Net Income to Common	487	1,168	1,951	1,976	599	(650)	115	1,216	2,421
EBITDA	1,755	2,320	3,271	3,361	2,238	325	2,541	4,658	6,522
EPS ex-Equity Income (KRW)	3,819	10,986	19,900	19,360	6,727	(13,478)	1,905	15,760	34,123
EPS - Total (KRW)	7,439	17,855	29,810	30,206	8,917	(9,934)	1,752	18,577	37,006
BPS (KRW)	194,064	219,885	246,524	276,815	295,583	323,373	345,545	363,723	399,933
Diluted WASO	65.4	65.4	65.4	65.4	65.4	65.4	65.4	65.4	65.4
Balance Sheet	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & Cash Equivalents	1,669	2,433	3,097	2,075	1,885	1,804	1,981	1,630	3,760
Accounts Receivables	2,178	2,525	3,350	3,815	5,570	3,079	3,307	4,274	4,893
Inventories	1,811	2,487	3,205	3,297	2,879	2,936	2,840	3,365	3,324
Other Current Assets	-	-	-	-	-	921	-	-	-
Current Assets	5,657	7,445	9,652	9,187	10,334	8,740	8,128	9,269	11,977
Investments	8,576	9,543	10,145	11,304	10,187	11,427	12,372	12,601	12,835
Property, plant & equipment	6,031	7,544	8,890	11,862	17,484	18,935	20,250	20,995	21,734
Intangible Assets/Goodwill	890	890	890	890	890	890	890	890	890
Other Non-current Assets	379	411	681	796	1,702	2,263	2,715	2,715	2,715
Non-Current Assets	15,877	18,388	20,606	24,852	30,263	33,515	36,227	37,201	38,174
Total Assets	21,534	25,833	30,258	34,039	40,597	42,255	44,355	46,471	50,151
Accounts Payable	869	1,272	2,192	2,119	906	1,015	1,037	1,363	1,697
Short-Term Borrowings	2,451	2,527	2,879	2,894	6,550	5,498	5,222	5,157	5,093
Other Short-Term Liabilities	1,663	2,662	2,936	3,506	3,400	3,282	3,792	3,792	3,792
Current Liabilities	4,984	6,461	8,007	8,519	10,856	9,795	10,050	10,312	10,581
Long-Term Borrowings	1,508	2,129	2,341	2,905	5,190	6,291	6,014	5,950	5,885
Other Long-Term Liabilities	1,684	2,046	2,692	2,708	2,985	2,599	3,000	3,000	3,000
Long-term Liabilities	3,192	4,175	5,033	5,613	8,174	8,890	9,014	8,950	8,885
Total Liabilities	8,175	10,637	13,040	14,132	19,030	18,685	19,064	19,262	19,466
Shareholders' Equity	13,359	15,197	17,218	19,907	21,567	23,570	25,291	27,209	30,685
Total Liabilities and Equity	21,534	25,833	30,258	34,039	40,597	42,255	44,355	46,471	50,151
Cash Flow Statement	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Net Income	631	1,250	2,039	2,066	576	(585)	437	1,918	3,476
Depreciation & Amortization	1,084	1,252	1,463	1,728	1,875	2,047	2,326	2,418	2,504
Investment Income	(293)	(596)	(844)	(853)	(169)	(359)	(281)	(229)	(234)
Changes in Net Working Capital	205	(620)	(623)	(631)	(2,550)	2,544	692	(1,167)	(244)
Other	322	889	605	(207)	131	(2,636)	85	-	-
Cash Flow From Operations	1,949	2,176	2,641	2,104	(138)	1,010	3,259	2,941	5,501
Capex	(1,572)	(2,180)	(2,518)	(4,254)	(6,600)	(3,330)	(3,163)	(3,163)	(3,243)
Other	(207)	231	(428)	149	1,680	747	(28)	-	-
Cash Flow From Investing	(1,778)	(1,950)	(2,946)	(4,105)	(4,920)	(2,583)	(3,191)	(3,163)	(3,243)
Cash Flow From Financing	241	583	629	903	5,544	1,109	217	(129)	(129)
Cash at Beginning of Period	1,148	1,559	2,368	2,692	1,593	2,080	1,615	1,793	1,441
Net Cash Flow	411	809	324	(1,099)	487	(465)	177	(351)	2,130
Adjustments	110	65	406	482	(195)	189	189	189	189
Cash at End of Period	1,669	2,433	3,097	2,075	1,885	1,804	1,981	1,630	3,760
Free cash flow	377	(4)	123	(2,151)	(6,738)	(2,320)	96	(222)	2,259

Source: Company data, Bernstein estimates and analysis

EXHIBIT 30: LG Chem financials outlook

Income Statement (KRW bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Revenue	30,077	42,655	51,865	55,658	48,699	45,932	52,782	64,738	77,552
COGS	(23,764)	(31,438)	(41,878)	(46,834)	(41,248)	(38,123)	(41,076)	(46,262)	(53,194)
Gross Profit	6,312	11,219	9,987	8,824	7,451	7,809	11,705	18,477	24,358
SG&A	(4,514)	(6,194)	(6,991)	(6,985)	(8,058)	(8,275)	(9,871)	(12,113)	(14,515)
Operating Profit	1,798	5,025	2,996	2,514	875	1,182	1,834	6,364	9,843
Finance income/(expense)	417	(271)	56	414	619	870	618	612	768
Investment Income	9	-	-	-	-	-	-	-	-
Other non-operating income/(expense)	1,155	137	(237)	(438)	(1,804)	(3,832)	(1,206)	(600)	(600)
Income Before Taxes	944	4,891	2,815	2,490	(310)	(1,780)	1,246	6,375	10,011
Income Tax Expense (Income)	368	937	619	436	(826)	(802)	669	1,594	2,503
Profit (loss) from discontinued operations	-	-	-	-	-	-	-	-	-
Net Income	682	3,954	2,195	2,054	516	(978)	577	4,782	7,508
Minority Interest/Preferred	221	609	350	716	1,207	(1)	366	2,531	2,939
Net Income to Common	461	3,345	1,845	1,338	(691)	(977)	211	2,250	4,569
EBITDA	4,110	7,870	6,350	6,471	5,547	6,535	7,969	12,615	16,151
EPS (KRW)	6,666	47,108	23,670	17,089	(8,826)	(13,259)	2,693	28,743	58,366
BPS (KRW)	271,643	326,861	480,759	522,936	613,173	639,199	631,645	692,721	788,624
Balance Sheet (KRW bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & Cash Equivalents	3,274	3,825	8,512	9,142	8,068	10,903	15,485	22,913	32,894
Accounts Receivables	5,298	6,387	8,957	8,083	8,166	6,725	7,982	9,995	12,143
Inventories	5,350	8,284	12,493	9,375	8,847	8,177	7,625	8,725	10,309
Other Current Assets	2,397	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Current Assets	16,320	19,496	30,962	27,600	26,081	26,805	32,093	42,633	56,345
Property, plant & equipment	20,567	26,823	32,353	42,643	58,190	59,960	62,760	63,350	63,883
Intangible Assets/Goodwill	2,280	2,682	3,235	4,264	5,819	5,996	6,276	6,335	6,388
Investments	309	309	309	309	309	309	309	309	309
Other Non-current Assets	1,912	1,824	1,114	2,650	3,459	7,992	8,896	8,896	8,896
Non-Current Assets	25,069	31,639	37,012	49,867	67,777	74,257	78,241	78,890	79,476
Total Assets	41,389	51,135	67,974	77,467	93,858	101,062	110,334	121,523	135,822
Accounts Payable	3,459	3,742	4,756	4,118	3,682	3,537	4,106	4,698	5,551
Short-Term Borrowings	1,549	3,477	3,804	7,078	7,621	11,738	14,976	17,884	20,853
Other Short-Term Liabilities	7,617	7,617	7,617	7,617	7,617	7,617	7,617	7,617	7,617
Current Liabilities	12,624	14,836	16,177	18,813	18,920	22,892	26,699	30,199	34,021
Long-Term Borrowings	8,637	11,289	12,160	14,850	19,755	22,074	24,397	27,305	30,274
Other Long-Term Liabilities	1,337	1,800	2,156	2,866	7,187	8,990	9,787	9,787	9,787
Long-term Liabilities	9,974	13,089	14,316	17,716	26,942	31,064	34,184	37,092	40,061
Total Liabilities	22,598	27,925	30,493	36,529	45,862	53,956	60,884	67,291	74,082
Shareholders' Equity	18,790	23,210	37,481	40,938	47,996	47,106	49,450	54,232	61,740
Total Liabilities and Equity	41,389	51,135	67,974	77,467	93,858	101,062	110,334	121,523	135,822
Net Debt	6,912	10,941	7,452	12,786	19,308	22,909	23,888	22,276	18,233
	-	-	-	-	-	-	-	-	-
Cash Flow Statement (KRW bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Net Income	682	3,954	2,195	2,054	516	(745)	577	4,782	7,508
Depreciation & Amortization	2,312	2,845	3,354	3,957	4,672	5,353	6,135	6,251	6,308
Changes in Net Working Capital	(875)	(3,738)	(3,646)	1,234	8	1,966	(136)	(2,521)	(2,879)
Other	3,895	2,453	(1,333)	291	1,817	1,661	929	-	-
Cash Flow From Operations	6,014	5,514	570	7,536	7,013	8,235	7,506	8,512	10,937
Capex	(5,300)	(7,100)	(9,822)	(14,301)	(14,924)	(13,310)	(7,497)	(6,900)	(6,894)
Other	(178)	1,746	593	1,175	1,416	1,629	671	-	-
Cash Flow From Investing	(5,478)	(5,354)	(9,229)	(13,126)	(13,508)	(11,681)	(6,826)	(6,900)	(6,894)
Borrow/Repay	1,772	4,580	1,199	5,963	5,447	6,437	5,562	5,815	5,938
Other	(826)	(4,457)	12,133	195	(556)	(194)	(1,820)	-	-
Cash Flow From Financing	946	123	13,332	6,158	4,891	6,243	3,742	5,815	5,938
Net Cash Flow	1,483	283	4,673	568	(1,604)	2,797	4,421	7,428	9,981
Cash at Beginning of Period	11,558	16,800	23,004	31,062	34,440	31,020	50,773	76,374	111,730
FX effect	(97)	559	64	76	303	1,357	161	-	-
Cash at End of Period	12,944	17,642	27,741	31,706	33,139	35,174	55,355	83,802	121,711

Source: Company data, Bernstein estimates and analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY**LG Energy Solution**

We value LGES using the DCF approach with a WACC of 9.6% and a terminal growth rate of 3%. Our DCF model is based on annual free cash flow forecasts until 2050, plus a terminal value estimate to capture the continuing value of the company. Our price target is KRW351,000.

LG Chem Ltd

We value LG Chem using a SOTP methodology to arrive at our KRW312,000 target price. We use the DCF approach with a WACC of 9.6% and a terminal growth rate of 2% to value the Energy Solutions business, which comprises ~70% of our SOTP-based target price. We value Petrochemicals business using 2025-27 P/E multiple of 10x, Advanced Materials business using a 2025 EV/Sales of 2.0x and Life Sciences business using a 2025-27 P/E of 50x.

Samsung SDI Co Ltd

We value Samsung SDI using the DCF approach with a WACC of 9.3% and a terminal growth rate of 3%. Our DCF model is based on annual free cash flow forecasts until 2050, plus a terminal value estimate to capture the continuing value of the company. Our price target is KRW520,000

RISKS**LG Energy Solution**

Key upside risks include 1) increasing market share with customer diversification which could increase sales, 2) rising margins with better costs control and raw material pass through and 3) improved battery quality which could lower the risks of recalls.

Key downside risks include 1) lower tax benefits from US IRA policy which LGES currently benefits from, 2) redistribution of tax credits with customers or partners, and 3) cost inflation in battery manufacturing.

LG Chem Ltd

Downside risks include: 1) increasing competition within the EV battery industry, 2) rising raw material costs which could put additional pressure on the EV value chain, 3) battery quality issues which could lead to battery recalls, and 4) a larger than expected conglomerate discount. In addition, oversupply of chemicals and weak demand could create sustained lower margins.

Key upside risks include: 1) divestment of the LGES stake; 2) faster growth in demand for EV batteries and related materials; and 3) margin expansion driven by improved cost management and effective raw material cost pass-through.

Samsung SDI Co Ltd

Samsung SDI's earnings growth depends on the adoption of electric vehicles and energy storage systems to boost battery revenues and profits. Any change in strategy by automakers, or lack of cost declines would reduce this upside. Other downside

risks include loss of sales to OEM switching battery suppliers. A reduction of US IRA subsidies would reduce benefits to Samsung SDI's profitability which is also a downside risk.

Key upside risks comprise: 1) higher-than-anticipated demand for ESS batteries, particularly driven by robust data center requirements; 2) increased EV battery demand from clients such as BMW following the launch of new models; and 3) accelerated commercialization advancements in solid-state battery technology

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors

on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

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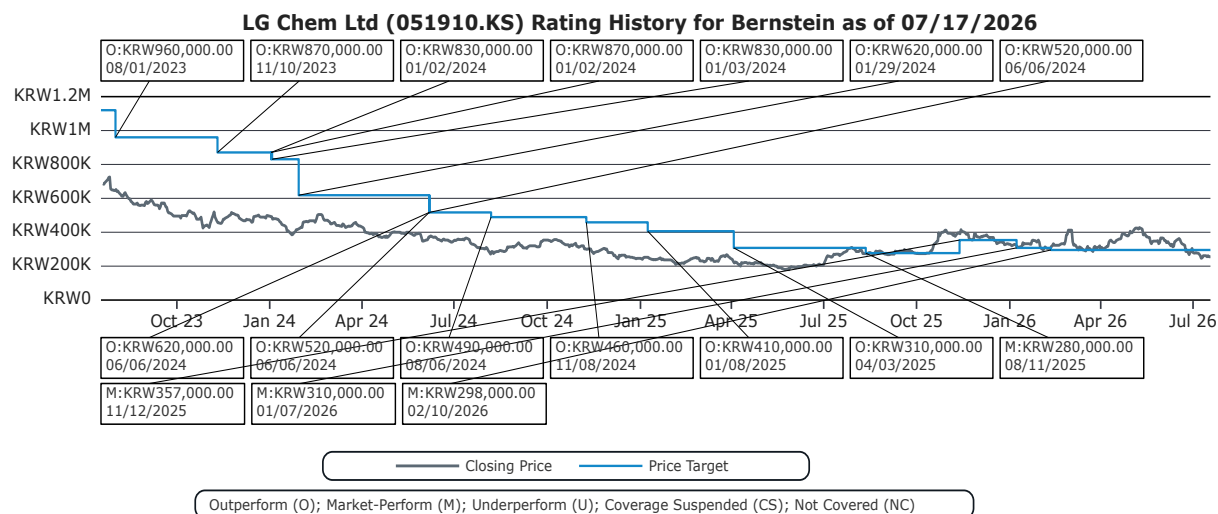
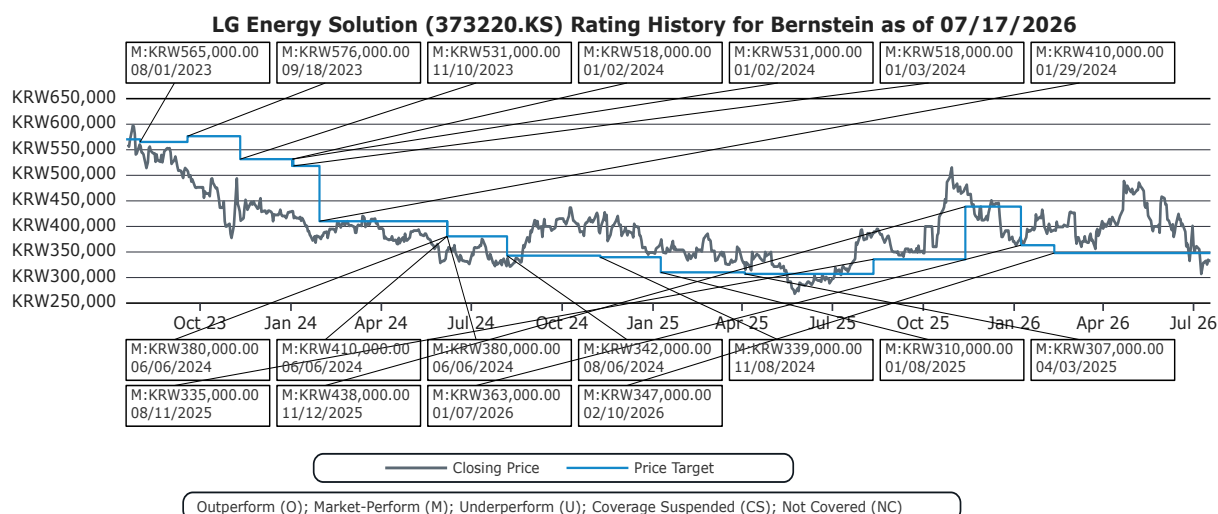
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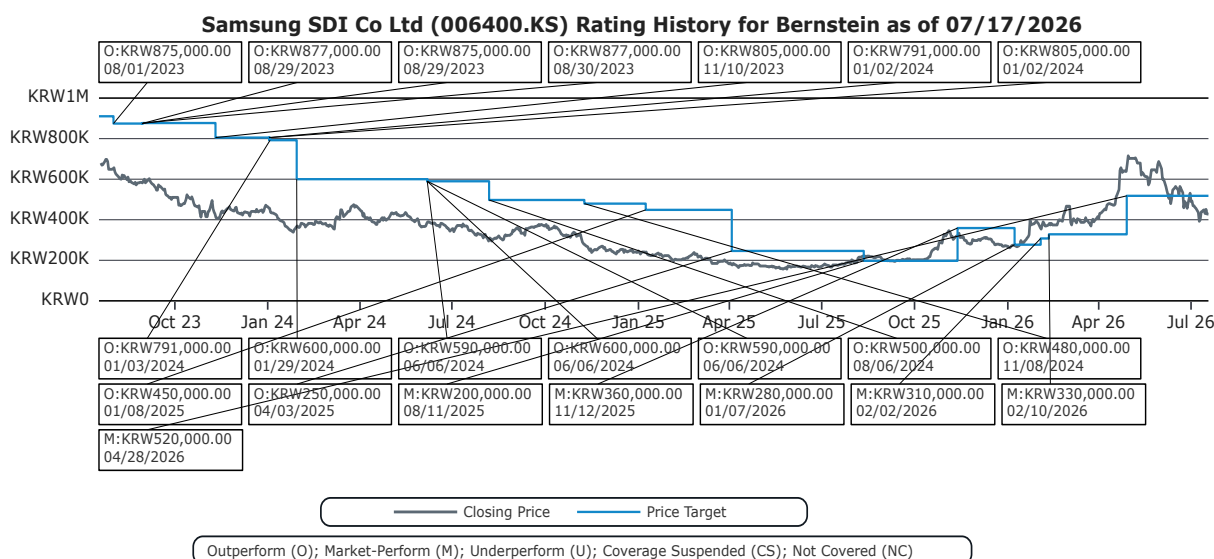
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